Element F:

Project Viability:

The aquaponic system project for a local middle school, designed to grow tilapia and root vegetables within a $1000 budget, is highly viable. This is supported by evidence that small-scale aquaponics can yield 62 kg of fish and 352 kg of produce with a net value of 151.3 €, far exceeding the project's budget constraints. Additionally, aquaponics aligns with educational goals, adheres to safety codes, and has a positive environmental impact, making it an excellent choice for the school setting. The project's design and budget are realistic, ensuring its success and sustainability.

Hazards/risks of the project that could fail and how to avoid them:

Risk	**Prevention**
The structure of the housing barrels could collapse from the water/soil weight	Make sure all metrics are stable and not volatile throughout the year. Consistent weight and correct/reasonable construction for that weight will make collapsing plant beds improbable.
The tilapia could die from the rest of a new location and travel	Ensure a quick and stress free travel, which can be bought from local and closer fisheries. The Tilapia are hardy enough to withstand new environments quickly, however to mitigate stress, we can acclimate them to the water temperature and feed them regularly.
Small leaks and evaporation loss over time could drain all the necessary water volume as time goes on without clear signs	Add sensors and measurements daily to be alerted when water levels are adequate. Add an inner lining and inspections throughout the whole aquaponic system to catch any leaks that may arise. Create a cool and dark environment for the water tank to limit/prevent evaporation significantly.
Children at the middle school may be rough with it and cause trouble for the fish and plants.	Gate off the area to only teacher overviewed sessions with classes. Make the exposed area of the location less open and protected with a temporary gate or barrier. General education of caution may help also to prevent the kids from ignorantly messing up the system.